Order Statistics and Their Functions

STA 312: Statistical Theory I

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Lecture Outline

1. Introduction
2. Distributions of Extremes
3. General Order Statistics
4. Special Cases & Functions

Definition of Order Statistics

The Setup

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a continuous population with PDF $f(x)$ and CDF $F(x)$.

If we arrange these observations in increasing order of magnitude:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$$

- $X_{(1)} = \min(X_1, \ldots, X_n)$ is the **1st Order Statistic**.
- $X_{(r)}$ is the **r-th Order Statistic**.
- $X_{(n)} = \max(X_1, \ldots, X_n)$ is the **n-th Order Statistic**.

Note: For continuous distributions, $P(X_i = X_j) = 0$, so we strictly have $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ almost surely.

Distribution of the Maximum, $X_{(n)}$

The CDF of the maximum is the probability that *all* observations are $\leq x$:

$$F_{X_{(n)}}(x) = P(X_1 \leq x, \ldots, X_n \leq x) = [F(x)]^n$$

The PDF of $X_{(n)}$

Differentiating with respect to x gives:

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x)$$

Distribution of the Minimum, $X_{(1)}$

The CDF of the minimum is $1 - P(X_{(1)} > x)$, which is the probability that at least one observation is $\leq x$:

$$F_{X_{(1)}}(x) = 1 - P(X_1 > x, \ldots, X_n > x) = 1 - [1 - F(x)]^n$$

The PDF of $X_{(1)}$

Differentiating gives:

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

The General r-th Order Statistic

The PDF of $X_{(r)}$ for $1 \leq r \leq n$ is given by:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!}[F(x)]^{r-1}[1-F(x)]^{n-r}f(x)$$

Heuristic Interpretation:

- One observation falls in the interval $(x, x+dx) \rightarrow f(x)dx$.
- $r-1$ observations fall below $x \rightarrow [F(x)]^{r-1}$.
- $n-r$ observations fall above $x \rightarrow [1-F(x)]^{n-r}$.
- Total permutations: $\frac{n!}{(r-1)!1!(n-r)!}$.

Special Case: Order Statistics from $U(0,1)$

Suppose $X_i \sim \text{Uniform}(0,1)$. Then $F(x) = x$ and $f(x) = 1$ for $0 < x < 1$.

The Beta Connection

Substituting into the general formula:

$$f_{X_{(r)}}(x) = \frac{n!}{(r-1)!(n-r)!} x^{r-1}(1-x)^{n-r}$$

This is the PDF of a **Beta Distribution** with parameters $\alpha = r$ and $\beta = n - r + 1$.

$$E[X_{(r)}] = \frac{r}{n+1}, \quad Var(X_{(r)}) = \frac{r(n-r+1)}{(n+1)^2(n+2)}$$

Functions of Order Statistics

In statistical practice, we often use functions of order statistics:

Function	**Formula**	**Statistical Use**
Sample Range	$R = X_{(n)} - X_{(1)}$	Measure of Dispersion
Sample Midrange	$M = \frac{X_{(1)}+X_{(n)}}{2}$	Measure of Centrality
Sample Median	$X_{(\frac{n+1}{2})}$ (odd n)	Robust Centrality

Joint PDF of $X_{(1)}$ and $X_{(n)}$

$$f_{X_{(1)},X_{(n)}}(x_1, x_n) = n(n-1)[F(x_n) - F(x_1)]^{n-2} f(x_1) f(x_n)$$

for $x_1 < x_n$.

Summary

- Order statistics depend on the rank of values, not just their magnitude.
- They are **not independent**, even if the parent sample was i.i.d.
- Their distributions are essential for:
  1. Non-parametric inference.
  2. Analysis of extremes (e.g., flood levels, stock market crashes).
  3. Characterizing the reliability of systems.

Thank You!

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Questions?